ANY TWO ALGEBRAIC CLOSURES OF A FIELD ARE ISOMORPHIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Uniqueness half of the algebraic-closure package; the existence half is lemma-map-into-algebraic-closure (worklist row 15, still pending as its own card).

Editorial note. The label lemma-algebraic-closures-isomorphic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-map-into-algebraic-closure (source lemma, worklist row 15): for any algebraic extension M of F there is a morphism of F-extensions M into a given algebraic closure
- Chapter lemma-algebraic-goes-up (source lemma, worklist row 93): images of algebraic elements under an F-homomorphism are algebraic
- Chapter lemma-algebraically-closed (worklist row 13): an algebraically closed field has no irreducible polynomials beyond the linear ones, hence admits no proper algebraic extension

Source reference. Original reference 09GF: sources/stacks-fields/pinned-original-source.tex, source lines 860–860. Prerequisite pointer invoked at 1280; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GU: sources/stacks-fields/pinned-original-source.tex, source lines 1240–1240. Prerequisite pointer invoked at 1277; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Any two algebraic closures of a field are isomorphic.*

Proof. Let F be a field. If M and $\overline{F}$ are algebraic closures of F, then there exists a morphism of F-extensions $\varphi : M \to \overline{F}$ by Lemma 09GU. Now the image $\varphi(M)$ is algebraically closed. On the other hand, the extension $\varphi(M) \subset \overline{F}$ is algebraic by Lemma 09GF. Thus $\varphi(M) = \overline{F}$. □

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